AI-POWERED PALMISTRY & TAROT INTELLIGENCE PLATFORM

Comprehensive Technical Specification, Architecture & Production Dossier

Production Status: Active & Live (SSL Enabled)	Version: 1.0.0 (Dockerized)
Live Web App: https://ai-palmistry-tarot-latest.onrender.com	**Docker Hub:** ponnarameshkumar/ai-palmistry-tarot:latest
Interactive API Docs: https://ai-palmistry-tarot-latest.onrender.com/docs	**GitHub Repo:** https://github.com/ponnarameshkumar099/AI-palmistry-tarot-platform

1. Executive Summary & Product Vision

The **AI-Powered Palmistry & Tarot Intelligence Platform** is an enterprise-grade multimodal artificial intelligence platform that unifies traditional self-reflection arts (Chiromancy, Tarot Arcana, and Natal Astrology) with state-of-the-art Computer Vision (CV), Machine Learning (ML), and psychometric analytics. The platform converts subjective esoteric readings into consistent, mathematical features, producing transparent career alignment scores and actionable 4-phase execution roadmaps.

Key capabilities include automated 21-landmark palm line segmentation, complete 78-card Major & Minor Arcana engine with interactive 3D physics, Scikit-Learn Random Forest career classification, ReportLab PDF report generation, and fully containerized Docker architecture deployed globally on Render.com with SSL/TLS encryption.

2. Multi-Role Authentication & Access Control (RBAC)

The platform implements strict Role-Based Access Control (RBAC) via FastAPI security dependencies and JWT tokens. Four distinct user roles are pre-configured for evaluation:

Role	Email	Password	Key Permissions & Scope
Seeker (User)	user@mystic.io	password123	Upload hand scans, draw 78-card tarot spreads, access ML roadmaps, export reports, book sessions.
Tarot Reader	reader@mystic.io	ReaderPass2026!	Review client spread history, annotate readings, configure custom spread geometries, client notes.
Consultant	consultant@mystic.io	ConsultantPass2026!	Manage appointment queues, provide holistic career synthesis, inspect biometric trends.
Administrator	admin@mystic.io	AdminPass2026!	System health telemetry, database backup/restore, user role governance, audit trails.

3. Core Intelligent Subsystems

3.1 Computer Vision Palmistry Studio

The CV Palmistry module extracts structural metrics from palm photographs using geometric landmark detection. It maps 21 anatomical anchors across wrist baselines, mount elevations, and metacarpophalangeal joints to evaluate four major flexion creases:

- **Heart Line:** Segmented across upper hypothenar to index index mount; evaluates emotional resilience, interpersonal empathy, and social boundary dynamics.
- **Head Line:** Measured for curvature and terminal descent; quantifies logical precision, strategic conceptualization, and creative reasoning.
- **Life Line:** Curved along thenar eminence; measures vitality baseline, constitutional stamina, and adaptability to operational disruption.
- **Fate (Saturn) Line:** Vertical axis calculation measuring autonomy, executive drive, leadership initiative, and career stability.
- **Responsive HUD Architecture:** Engineered with full-screen width efficiency, displaying the camera canvas on the left while dedicating an adjacent right-side HUD box to percentage clarity scores and hand confidence metrics.

3.2 Complete 78-Card Tarot Intelligence Engine

Unlike standard applications limited to 22 Major Arcana, this platform implements the entire canonical 78-card deck across Major and Minor Arcana:

- **22 Major Arcana:** Fundamental archetypal life cycles from The Fool (0) to The World (XXI) with deep astrological correspondences.
- **56 Minor Arcana:** 14 cards per elemental suit (Ace through 10, Page, Knight, Queen, King):

- *Wands (Fire):* Enterprise, vision, passion, kinetic energy, creative drive.
- *Cups (Water):* Intuition, emotional intelligence, relationships, empathy.
- *Swords (Air):* Mental clarity, strategic communication, conflict analysis.
- *Pentacles (Earth):* Tangible assets, career execution, material stability, skill mastery.

- **3D Interactive Card Fan:** Dynamic card fanning with physics elevation, reversible upright/inverted orientation logic, and multi-card spread configurations (Single Card, 3-Card Timeline, 5-Card Celtic Cross).

3.3 Machine Learning Career Guidance & Multimodal Synthesis

The recommendation engine applies a Scikit-Learn Random Forest Classifier trained on multidimensional psychometric feature sets. It computes a unified 0–100% Career & Life Alignment score and generates a structured 4-phase execution roadmap:

- **Phase 1: Foundation (0–30 Days):** Skill audits, cognitive habit recalibration, core technical immersion.
- **Phase 2: Strategic Acceleration (30–60 Days):** Portfolio formulation, stakeholder networking, high-impact prototypes.
- **Phase 3: Executive Positioning (60–90 Days):** Leadership expansion, authority positioning, contract negotiation.
- **Phase 4: Long-Term Legacy:** Thought leadership, ethical capital preservation, sustainable mentorship.

4. Docker Containerization & Cloud Infrastructure

The application is engineered to adhere to 12-factor cloud-native principles, packaging backend logic, static frontend assets, and ML model binaries into an immutable container.

Component	Implementation Specification
Base Image	python:3.11-slim (minimal Debian bookworm runtime, minimal attack surface)
Dependencies	FastAPI, Uvicorn, Scikit-learn, ReportLab, Pydantic, NumPy, Pillow, Chart.js, Tailwind CDN
Dynamic Cloud Port	run.py inspects os.environ.get('PORT', 8000), ensuring zero-configuration cloud hosting across Render, AWS, and Railway
Healthcheck	HEALTHCHECK CMD curl -f http://localhost:8000/docs || exit 1 running every 30s
Docker Hub Image	ponnarameshkumar/ai-palmistry-tarot:latest (public cloud container registry)
Cloud Production	Render.com Web Service with automated HTTPS, TLS 1.3, and zero-downtime rolling deploys

5. Local & Cloud Execution Guide

The project provides multiple execution modalities suitable for local development, containerized testing, or remote cloud deployment:

Option A: 1-Click Windows Launch (Local Docker)

Double-click docker-run.bat in the project root. The script verifies Docker Desktop daemon availability, compiles the image with docker compose up --build -d, and opens http://localhost:8000 in your default browser.

Option B: Terminal Command (Any OS with Docker)

```
# Build and start container in background
docker compose up --build -d

# View live stream logs
docker logs -f palmistry_tarot_platform

# Run directly from Docker Hub without source code
docker run -d -p 8000:8000 ponnarameshkumar/ai-palmistry-tarot:latest
```

6. RESTful API Router Catalog

Router Prefix	Key Endpoints	Functional Description
/api/auth	/login, /register, /me, /demo-users	OAuth2 authentication, JWT generation, demo role switching.
/api/palm	/analyze, /analyze-json, /presets	Multipart scan intake, landmark segmentation, preset repository.
/api/tarot	/deck, /spreads, /read, /history	78-card catalog, spread layouts, card shuffling engine.
/api/readings	/synthesize, /oracle-ask, /history	Cross-discipline synthesis, conversational oracle AI consultation.
/api/intelligence	/ml-personality-predict, /ml-career-roadmap	Scikit-learn random forest career roadmap scoring.
/api/export	/pdf, /json	Server-side ReportLab PDF dossiers & raw JSON reports.
/health	/health	Cloud container uptime, active modules, and subsystem telemetry.

7. Future Scope & System Evolution

- **On-Device Edge Neural Inference:** Deploying CoreML and TensorFlow Lite models for native iOS/Android camera scanning with zero server latency.
- **Real-time Generative Voice Oracle:** Incorporating streaming neural voice synthesis (TTS) to deliver spoken oracle consultations.
- **Zero-Knowledge Biometric Privacy:** Cryptographic zero-knowledge proofs ensuring raw biometric palm prints never leave the user's local device.